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SUBJECT CODE NO: - RNP-8001**FACULTY OF SCIENCE AND TECHNOLOGY****F.E. (Group A & B) (NEP) Sem - I****Examination November / December 2025 (To be held in February-2026)****Applied Mathematics-I****[Time: 3:00 Hours]****[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A**Q1 Solve the following questions:****20**

- 1) A circuit containing resistance R and inductance L in series with voltage source E then the D.E for current is:
 - a) $Li + R \frac{di}{dt} = E$
 - b) $L \frac{di}{dt} + Ri = E$
 - c) $Li - R \frac{di}{dt} = E$
 - d) $Li + R \frac{di}{dt} = 0$
- 2) The necessary & sufficient condition for D.E. to be exact D.E. is:
 - a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
 - b) $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$
 - c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
 - d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$
- 3) The solution of D.E. $xy(1 + xy^2) \frac{dy}{dx} = 1$ is:
 - a) $x + y^2 - 2 = ce^x$
 - b) $\frac{1}{x} + y^2 - 2 = ce^{-\frac{y^2}{2}}$
 - c) $xy^2 - 2 = ce^x$
 - d) $\frac{1}{x} + 2y^2 = ce^{-\frac{y^2}{2}}$
- 4) The rate of change of velocity is given by:
 - a) $\frac{dv}{dt}$
 - b) $\frac{dv}{dx}$
 - c) $a \frac{dv}{dx}$
 - d) $m \frac{dv}{dx}$
- 5) If $x = e^\theta \cos \phi$ $y = e^\theta \sin \phi$ then $\frac{\partial(x,y)}{\partial(\theta,\phi)}$ is:
 - a) $x + y$
 - b) $x^2 + y^2$
 - c) e^θ
 - d) 1
- 6) The stationary points of the function $f(x, y) = x^3 + y^3 - 3axy$ are:
 - a) $x = 0, y = a$
 - b) $x = a, y = 0$
 - c) $x = a, y = a$
 - d) $x = a, y = -a$

- b) Divide 24 into three parts such that the continued product of the first, square of the Second & cube of the third is maximum **05**

Q6 a) Expand $\log x$ in powers of $(x - 4)$ **05**

- b) Evaluate : $\lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{1}{\log(x-1)} \right]$ **05**

Q7 a) A constant electromotive force E volts is applied to a circuit containing a constant resistance R ohms in series and a constant inductance L henries. If the initial current is zero show that the current builds up to half of its theoretical maximum in $\frac{(L \log 2)}{R}$ seconds **05**

- b) A body of mass m , falls from rest under gravity in a fluid whose resistance to motion at any instant is mK times its velocity, where K is a constant. Find the terminal velocity of the body & also the time to acquire one half of its limiting speed. **05**